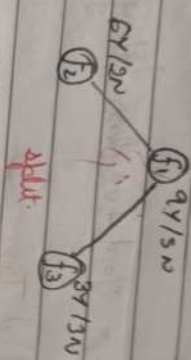


# Decision Tree Information Gain

① Entropy  $\rightarrow$  split purity.



{ The split will be going on until 2 values are got a pure output. }

$H(S) = -P_{+} \log_2 P_{+} - P_{-} \log_2 P_{-}$  (leaf node)

$H(S_2) = -\frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8}$

$\rightarrow 6+2 = 8$

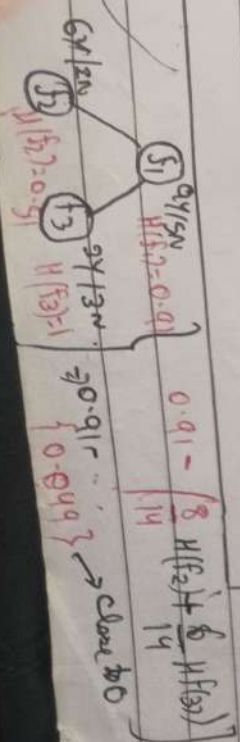
$\Rightarrow 0.91$

$0 < H(S) < 1$

• To find out the number of splits needed we use,

## Information Gain

Gain(S, A) =  $H(S) - \sum_{v \in V(A)} H(S_v)$



$0.91 - \left( \frac{8}{14} H(S_2) + \frac{6}{14} H(S_3) \right)$

$\Rightarrow 0.91 - \dots$

$\rightarrow 0.049 \rightarrow$  close to 0

$$\therefore \text{Gain}(S, f_1) \Rightarrow 0.049 \approx 0$$

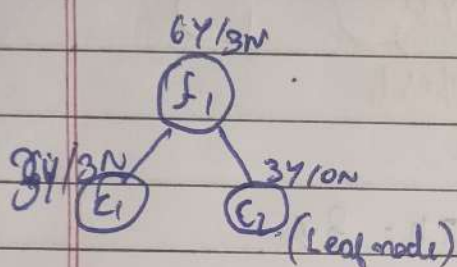
thus, it is pure.

Lecture - 39

## Gini Impurity DT

① Entropy  $\{ H(S) = -P_+ \log_2 P_+ - P_- \log_2 P_- \}$

## ② Gini Impurity



$$= 1 - \sum_{i=1}^n (p_i)^2$$

Probability

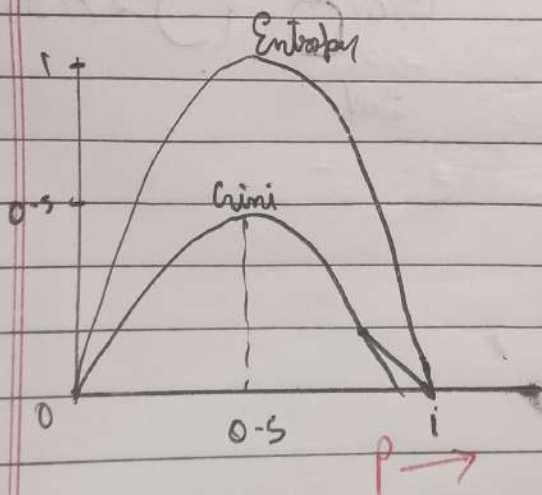
$$\Rightarrow 1 - [(P_+)^2 + (P_-)^2]$$

$$\Rightarrow 1 - \left[ \left( \frac{3}{6} \right)^2 + \left( \frac{3}{6} \right)^2 \right]$$

$$1 - [0.25 + 0.25]$$

$$1 - 0.5$$

$$\Rightarrow 0.5$$



When  $P_+$  is increasing

Entropy will inc.

& goes till 1

Whereas  $P_+$  is inc.

Gini imp. inc.

goes till  $\rightarrow 0.5$ .

$$\left\{ \begin{array}{l} 0 < En < 1 \\ 0 < GI < 0.5 \end{array} \right\}$$

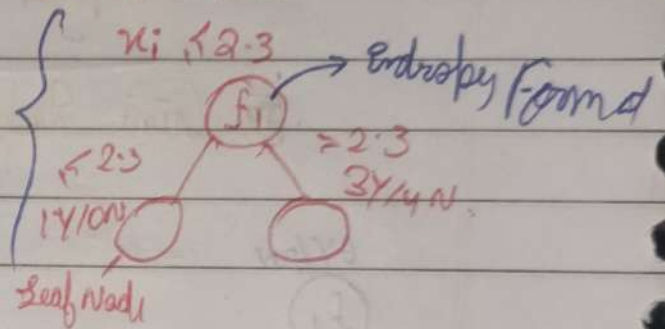
# Secture - 40

## Decision tree split for Numerical

| $f_i$ | o/p |
|-------|-----|
| 2.3   | Yes |
| 3.6   | Yes |
| 4     | No  |
| 5.2   | No  |
| 6.7   | Yes |
| 8.9   | No  |
| 10.5  | Yes |
| 14.2  | No  |

① Sorting all values

②  $T_h = 2.3$



$T_h = 3.6$

$x_i < 3.6$

